THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability Theory

Tutorial 13 Solution
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In this solution, we use p_{ij}^n to denote the probability $P(X_n = j \mid X_0 = i)$. Please distinguish it from the n-th power of p_{ij}, which will be denoted as $(p_{ij})^n$.

1. Show that the stationary probabilities for the Markov chain having transition probabilities p_{ij} are also the stationary probabilities for the Markov chain whose transition probabilities q_{ij} are given by
$$q_{ij} = p_{ij}^k$$
for any specified positive integer k.

   Solution: Denote the Markov chains as $\{X_n\}$ and $\{Y_n\}$, respectively, for the transition probabilities p_{ij} and q_{ij}. Let S be the state space. We first prove by mathematical induction that
$$q_{ij}^n = p_{ij}^{kn}, \quad \forall n \in \mathbb{N}.$$
(Be very careful about the notation here: p_{ij}^{kn} is the kn-step transition probability from i to j in $\{X_n\}$, while q_{ij}^n is the n-step transition probability from i to j in $\{Y_n\}$.) When $n = 1$, the above equation holds by the definition of q_{ij}. Assume that the above equation holds for some $n \in \mathbb{N}$. By Kolmogorov-Chapman equation, we have
$$q_{ij}^{n+1} = \sum_{l \in S} q_{il}^n q_{lj} = \sum_{l \in S} p_{il}^{kn} p_{lj}^k = p_{ij}^{k(n+1)}.$$
This completes the induction step.

   Now it follows that $\{q_{ij}^n\}_{n \geq 0}$ is a subsequence of $\{p_{ij}^n\}_{n \geq 0}$, so we have $\lim_{n\to\infty} q_{ij}^n = \lim_{n\to\infty} p_{ij}^n = \pi_j$, which is independent of i.

2. Prove that for any finite aperiodic Markov chain, there exists $N \in \mathbb{N}$ such that $p_{ii}^N > 0$ for every state i.

   Hint: Let $I_i = \{n \geq 1 \ : \ P_{ii}^n > 0\}$. We know that $\gcd(I_i) = 1$ and for any $m, n \in I_i$, $m+n \in I_i$. The question is equivalent to prove there exists $N \in \mathbb{N}$ such that $n \in I_i$ for all $n > N$. You may also make use of the fact that if $\gcd(a, b) = 1$, there exists $x, y \in \mathbb{Z}$ such that $ax + by = 1$.

   Solution: Fix any arbitrary state i. We first show that there exists a finite subset $S_i \subseteq I_i$ such that $\gcd(S_i) = 1$. Take any $n_0 \in I_i$. For each $d \in \mathbb{N}$ such that $d \mid n_0$, there exists $n_d \in I_i$ such that d does not divide n_d. Taking $S_i = \{n_0\} \cup \{n_d : d \mid n_0\}$, it is clear that S_i is finite, $S_i \subseteq I_i$ and $\gcd(S_i) = 1$. Denote $k := |S_i|$ and relabel the elements in S_i as $S_i = \{n_1, n_2, \ldots, n_k\}$.

   We now show that every sufficiently large n is a linear combination of elements in S_i with nonnegative integer coefficients. WLOG, let n_1 be the smallest element in S_i. Since $\gcd(S_i) = 1$, there exists $r_1, \ldots, r_k \in \mathbb{Z}$ such that $\sum_{j=1}^k r_j n_j = 1$. For each $j = 1, \ldots, k$, let $a_j = n_1 |r_j| \in \mathbb{Z}$, and denote
$$N_i = \sum_{j=1}^k a_j n_j.$$

Thus, for any $n \geq N_i$, there exists $b, t \in \mathbb{Z}$ such that $n - N_i = bn_1 + t$, where $b \geq 0$ and $t \in [0, n_1 - 1]$. In particular, we have

$$n = \sum_{j=1}^{k} a_j n_j + bn_1 + t\left(\sum_{j=1}^{k} r_j n_j\right) = (a_1 + tr_1 + b)n_1 + \sum_{j=2}^{k}(a_j + tr_j)n_j. \tag{1}$$

Since $a_j = n_1|r_j| > |tr_j|$, all the coefficients on the right hand side of (1) are nonnegative integers.

By the definition of I_i and the Chapman-Kolmogorov equation, we know that for any $m, n \in I_i$, $m + n \in I_i$. An inductive argument using the decomposition (1) implies that $n \in I_i$ for all $n \geq N_i$. In particular, we have $P_{ii}^n > 0$ for all $n \geq N_i$.

Finally, since the Markov chain is finite, we can take $N = \max\{N_i \ : \ i \text{ is a state}\} \in \mathbb{N}$, which is the required number.

3. In the gambler's ruin problem, suppose that each game has probability p of winning and $q = 1-p$ of losing. Assume that the gambler's fortune is presently i, and suppose that we know that the gambler's fortune will eventually reach N (before it goes to 0). Given this information, show that the probability p that he wins the next game is

$$P(X_{n+1} = i+1 \mid X_n = i, \lim_{m\to\infty} X_m = N) = \begin{cases} \frac{p[1-(q/p)^{i+1}]}{1-(q/p)^i}, & \text{if } p \neq \frac{1}{2}, \\ \frac{i+1}{2i}, & \text{if } p = \frac{1}{2}. \end{cases}$$

Solution: Denote P_i as the unconditional probability that the gambler's fortune will eventually reach N starting from state i. Recall from the lecture notes that

$$P_i = \begin{cases} \frac{1-(q/p)^i}{1-(q/p)^N}, & \text{if } p \neq \frac{1}{2}, \\ \frac{i}{N}, & \text{if } p = \frac{1}{2}. \end{cases}$$

We thus have

$$\begin{aligned} & P(X_{n+1} = i+1 \mid X_n = i, \lim_{m\to\infty} X_m = N) \\ =& \frac{P(\lim_{m\to\infty} X_m = N \mid X_n = i, X_{n+1} = i+1) \cdot P(X_{n+1} = i+1 \mid X_n = i)}{P(\lim_{m\to\infty} X_m = N \mid X_n = i)} \\ =& \frac{P(\lim_{m\to\infty} X_m = N \mid X_{n+1} = i+1) \cdot P(X_{n+1} = i+1 \mid X_n = i)}{P(\lim_{m\to\infty} X_m = N \mid X_n = i)} \\ =& \frac{P_{i+1}P_{i,i+1}}{P_i} \\ =& \begin{cases} \frac{p[1-(q/p)^{i+1}]}{1-(q/p)^i}, & \text{if } p \neq \frac{1}{2}, \\ \frac{i+1}{2i}, & \text{if } p = \frac{1}{2}. \end{cases} \end{aligned}$$

4. For the Markov chain with states $\{1, 2, 3, 4\}$ whose transition probability matrix $\mathbf{P}$ is as specified as below, find f_{i3} and s_{i3} for $i = 1, 2, 3$.

$$\mathbf{P} = \begin{bmatrix} 0.5 & 0.2 & 0.2 & 0.1 \\ 0.2 & 0.2 & 0.2 & 0.4 \\ 0.1 & 0.2 & 0.2 & 0.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solution: Clearly, the states $1, 2, 3$ are transient and 4 is recurrent. Hence

$$\mathbf{P}_T = \begin{bmatrix} 0.5 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 \\ 0.1 & 0.2 & 0.2 \end{bmatrix}$$

and

$$S = (I - \mathbf{P}_T)^{-1} = \begin{bmatrix} 0.5 & -0.2 & -0.2 \\ -0.2 & 0.8 & -0.2 \\ -0.1 & -0.2 & 0.8 \end{bmatrix}^{-1} = \begin{bmatrix} 2.5 & 5/6 & 5/6 \\ 0.75 & 19/12 & 7/12 \\ 0.5 & 0.5 & 1.5 \end{bmatrix}.$$

Since $f_{i3} = \dfrac{s_{i3} - \delta_{i3}}{s_{33}}$, we have $f_{13} = \frac{5}{9}$, $f_{23} = \frac{7}{18}$ and $f_{13} = \frac{1}{3}$.

5. Let $\{X_n\}_{n\geq 0}$ be a branching process defined by $X_{n+1} = \sum_{i=1}^{X_n} \xi_i^{(n+1)}$, where $\{\xi_i^{(n)}\}$ are i.i.d. random variables independent of X_0. If $X_0 = 1$, prove that $\lim_{n\to\infty} P(X_n = 0 \mid X_0 = 1)$ is the smallest positive number satisfying the following equation in s:

$$s = \sum_{j=0}^{\infty} p_j s^j,$$

where $p_j = P(\xi_i^{(n)} = j)$.

Hint: Use Mathematical Induction to prove a relation between any solution s and $P(X_n = 0 \mid X_0 = 1)$.

Solution: For the clarity of presentation, we omit the condition $X_0 = 1$ in the solution and write $P(X_n = 0)$ instead of $P(X_n = 0 \mid X_0 = 1)$, which shall not cause confusion.

Let $P(n)$ be the proposition: If s is a solution of $s = \sum_{j=0}^{\infty} p_j s^j$, then $s \geq P(X_n = 0)$. We prove that $P(n)$ is true for all $n \in \mathbb{N}$ by mathematical induction. Clearly we have $P(X_1 = 0) = p_0 \leq p_0 + \sum_{j=1}^{\infty} p_j s^j = s$, which implies $P(1)$. Now assume that $P(k)$ is true for all $1 \leq k \leq n$ for some $n \in \mathbb{N}$. We thus have

$$s = \sum_{j=0}^{\infty} p_j s^j \geq \sum_{j=0}^{\infty} p_j (P(X_k = 0))^j = P(X_{k+1} = 0), \tag{2}$$

where the last equality can be obtained by conditioning on the value of X_1 and using the independence of the number of offspring produced by different individuals. This proves $P(n+1)$ and hence completes the induction.

It now suffices to show that $\lim_{n\to\infty} P(X_n = 0)$ is a solution of $s = \sum_{j=0}^{\infty} p_j s^j$. Since the state 0 is absorbing, we have $P(X_n = 0) \leq P(X_{n+1} = 0)$ for all $n \in \mathbb{N}$. Thus $\{P(X_n = 0)\}$ is an increasing sequence bounded above by 1, hence $\lim_{n\to\infty} P(X_n = 0)$ exists and is bounded in $[0, 1]$. Taking the limit on (2), we have

$$\lim_{k\to\infty} \sum_{j=0}^{\infty} p_j (P(X_k = 0))^j = \lim_{k\to\infty} P(X_{k+1} = 0) = \lim_{k\to\infty} P(X_k = 0).$$

It can be justified that

$$\lim_{k\to\infty} \sum_{j=0}^{\infty} p_j (P(X_k = 0))^j = \sum_{j=0}^{\infty} p_j \left(\lim_{k\to\infty} P(X_k = 0) \right)^j. \tag{3}$$

Thus, $\lim_{n\to\infty} P(X_n = 0)$ is a solution of $s = \sum_{j=0}^{\infty} p_j s^j$.

Remark: The equation (3) can be justified by showing that $h(s) := \sum_{j=0}^{\infty} p_j s^j$ is continuous on $[0, 1]$. This is a consequence of Abel's theorem and the fact $\sum_{j=0}^{\infty} p_j = 1$. Indeed, one can even state that $h(s)$ is differentiable in a neighbourhood of $s = 1$, since $h(s) = \psi(\ln(s))$ with ψ being the moment generating function of $\xi_i^{(n)}$. It is known that ψ is continuous and differentiable on an open neighbourhood of 0, so is the function $h(s)$ at $s = 1$.

Note that $h'(s) = \sum_{j=0}^{\infty} j p_j s^{j-1}$, $h(0) = p_0 \geq 0$, and $h(1) = \sum_{j=0}^{\infty} p_j = 1$. It follows that $h'(1) = \mathbb{E}[\xi_i^{(n)}]$ and h is increasing and convex on $[0, 1]$. By the above facts, if $\mathbb{E}[\xi_i^{(n)}] \leq 1$, the equation $s = \sum_{j=0}^{\infty} p_j s^j$ has only one solution in $[0, 1]$, which is $s = 1$. In this case, we have $\lim_{n\to\infty} P(X_n = 0) = 1$. If $\mathbb{E}[\xi_i^{(n)}] > 1$, the equation $s = \sum_{j=0}^{\infty} p_j s^j$ has two distinct solutions in $[0, 1]$. In this case, we have $\lim_{n\to\infty} P(X_n = 0) < 1$. This characterizes the ultimate extinction probability of any branching process.